

Lab Workshop on Embedded C and ARM Cortex Microcontrollers

Day3: Lab Experiments: C Programming

Topic: Datatypes and range mapping

Question1:

- 1. Print the size of short int (format specifier is %ld)**
- 2. Declare a short int variable(var1) with negative value and print its value using format specifiers of signed short int(%hi), unsigned short int(%hu) and unsigned int(%u)**
- 3. Declare another short int variable(var2) with out of range value and observe you are getting warning or error understand the warning or error. Incase of warning print the var2 print its value using format specifiers of signed short int(%hi), unsigned short int(%hu) and unsigned int(%u)**

Result:

size of short int is 2 bytes

--- with in range values ---

var1 value declared using short int is -3

var1 value using unsigned short int is 65533

var1 value using unsigned int is 4294967293

--- Overflow conversion: out of bound values ---

var2 value using short int is 3

var2 value using unsigned short int is 3

var2 value using unsigned int is 3

Topic: Storage Classes

Question2:

- 1. Declare a global variable(int glbr_var = 20) and declare a function(func) print the global variable in the main() and func()**
- 2. Declare a local variable (int lcl_var = 25;) in main function and try to print its value in main() and func() and if any errors identify why?**
- 3. Declare a local variable with same name as global variable and assign different value in main() try to print its value in main() and observe the order of priority for variable scope**

Result:

the global variable value in main function is 20

the local variable value declared in main function is 25

the global variable value in main function is 52
the global variable value in function definition is 20

Question3:

Declare a function(func()) and create a static variable(count) in func() and print its value and increment the count value, call the func() multiple times and observe the functionality of static storage class.

Result:

the static value is 0
the static value is 1
the static value is 2

Question4:

Declare a global variable(num) in file1 and declare a function(func()) in file 2 using extern call the func() declared @file2 from file1 and using extern print the global variable declared @ file1 in func() of file2.

Result:

called the func() in file2 from file1
the variable value declared in file1 is 45

Topic: Conditional Structures

Question5:

write a switch-case program for displaying the day of the week (eg: Monday for case 2) by providing the input from terminal.

Note: write all the programs using text editor (nano or vi) and compile using terminal.



```
int main()
{
    printf(" size of short int is %li bytes \n",sizeof(short int));

    printf(" \n --- with in range values --- \n");

    short int var1 = -3;

    printf(" var1 value declared using short int is %hi \n" , var1);
    printf(" var1 value using unsigned short int is %hu \n" , var1);
    printf(" var1 value using unsigned int is %u \n" , var1);

    printf(" \n --- Overflow conversion: out of bound values --- \n");

    short int var2  = 65539;

    printf(" var2 value using short int is %hi \n" , var2);
    printf(" var2 value using unsigned short int is %hu \n" , var2);
    printf(" var2 value using unsigned int is %u \n" , var2);

    printf(" ----- \n");

    return 0;

}
```



Edit



Fit to Screen

Find and
Replace

Tools



Edit on PC



```
#include <stdio.h>
```

```
int gbl_var = 20;
```

```
void func(){
```

```
printf(" the global variable value in function definition is %d \n",  
gbl_var);
```

```
// printf("\n the local variable value declared in main function is  
%d ", lcl_var);    // results an error // because lcl_var is out of  
scope
```

```
}
```

```
int main()
```

```
{
```

```
printf(" the global variable value in main function is %d \n",  
gbl_var);
```

```
int lcl_var = 25;
```

```
printf(" the local variable value declared in main function is %d  
\n", lcl_var);
```

```
int gbl_var = 52;
```

```
printf(" the global variable value in main function is %d \n",  
gbl_var);
```

```
func();
```

```
return 0;
```

```
}
```

```
#include <stdio.h>
```

```
int func(){
```

```
static int count;
```

```
printf(" the static value is %d \n",count);
```

```
count++;
```

```
}
```

```
int main()
```

```
{
```

```
func();
```

```
func();
```

```
func();
```

```
return 0;
```

```
}
```

```
#include <stdio.h>
```

```
int num = 10;
```

```
extern void func();
```

```
int main()
```

```
{
```

```
num = 45;
```

```
func();
```

```
return 0;
```

```
}
```

```
#include <stdio.h>
```

```
extern int num;
```

```
void func(){
```

```
printf(" called the func() in file2 from  
file1 \n");
```

```
printf(" the variable value declared in  
file1 is %d \n", num);
```

```
}
```

```
#include <stdio.h>

int main()
{

    int day_num;

    /* Input week number from user */
    printf("Enter day number to display week starting from Monday \n (eg: Wednesday = 3, sunday =7 ) \n");
    scanf("%d", &day_num);

    switch(day_num)
    {
        case 1:
            printf("The relevant day is: Monday");
            break;
        case 2:
            printf("The relevant day is: Tuesday");
            break;
        case 3:
            printf("The relevant day is: Wednesday");
            break;
        case 4:
            printf("The relevant day is: Thursday");
            break;
        case 5:
            printf("The relevant day is: Friday");
            break;
        case 6:
            printf("The relevant day is: Saturday");
            break;
        case 7:
            printf("The relevant day is: Sunday");
            break;
        default:
            printf("Please enter day number between 1 to 7.");
    }

    printf("\n");

    return 0;
}
```